Checklist for standardized reporting on hearing loss in case of mitochondrial disease

Roesch et al. – Mitochondrial disease and hearing loss in children – a systematic review

This checklist represents a proposal for future reporting, based on experience acquired during conduction of a systematic literature review on this topic. The purpose of this document is to ensure a minimum standard of reporting, concerning disease and audiometric investigations in pediatric patients.

Definition and description of disease

Qualitative and quantitative results to be reported:

- Genotype, including gene affected, variant identified, mode of inheritance and OMIM ID – if applicable
- Clinical phenotype, including presence or absence of symptoms other than hearing loss, affecting one or multiple organs

Definition and description of audiometric testing

Proposed investigations:

- Performance on both ears/sides separately
- Tympanometry including stapedial reflexes
- Auditory brainstem response (ABR) measurement, including use of separate recordings with monopolar (rarefaction/condensation) stimuli for detection of CM in the region of the stimulus.
- Otoacoustic emission (OAE) measurement

Qualitative and quantitative results to be reported:

- Separate reporting for each ear/side
- Qualitative reporting on tympanometry and stapedial reflexes, including level of stapedial reflex thresholds
- Qualitative reporting on OAE measurement
- Quantitative reporting on ABR measurement, including stimulus levels given in decibel, absolute and interpeak latency times given in milliseconds and amplitudes for different levels in comparison to age-adapted normative values
- Qualitative reporting on ABR measurement, concerning detection of cochlear microphonics